INTRODUCTION

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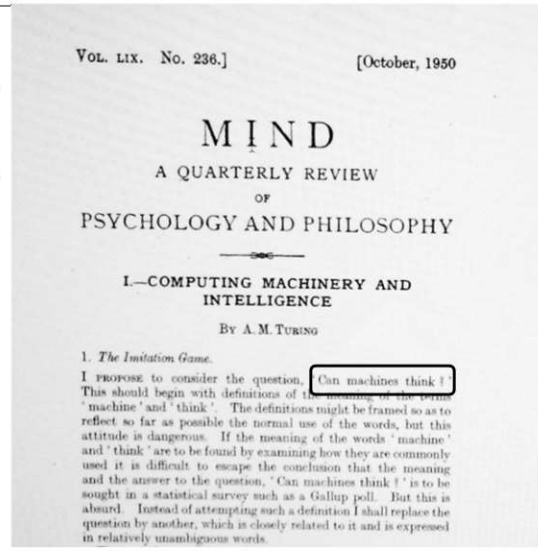
WHAT IS (ARTIFICIAL) INTELLIGENCE?

Artificial **I**ntelligence (AI)

Alan Turing's Perspective

1912 - 1954

Can machine think?



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MIND

A QUARTERLY REVIEW

OF

PSYCHOLOGY AND PHILOSOPHY

I.—COMPUTING MACHINERY AND INTELLIGENCE

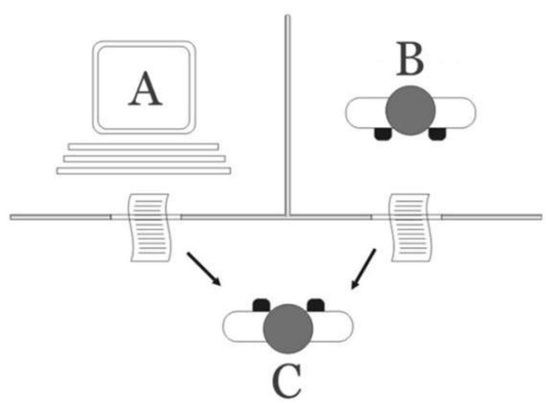
BY A. M. TURING

1. *The Imitation Game.*

I PROPOSE to consider the question, 'Can machines think ?' This should begin with definitions of th[illegible] 'machine' and 'think'. The definitions might be framed so as to reflect so far as possible the normal use of the words, but this attitude is dangerous. If the meaning of the words 'machine' and 'think' are to be found by examining how they are commonly used it is difficult to escape the conclusion that the meaning and the answer to the question, 'Can machines think ?' is to be sought in a statistical survey such as a Gallup poll. But this is absurd. Instead of attempting such a definition I shall replace the question by another, which is closely related to it and is expressed in relatively unambiguous words.

Turing test (Imitation Game)

- C (human evaluator) interacts with A (machine) and B (human) through natural languages.
- C tries to tell which one is the machine.
- If C cannot reliably tell, then the machine wins.



https://en.wikipedia.org/wiki/Turing_test

Different Goals of AI

Behave like human

Neural Art

Speech synthesis

Chatbot

Behave rationally

AutoTrader

AlphaGo

Healthcare

https://deepmind.google/research/projects/alphago/

Different Goals of AI

- Disciplines studying "**how human thinks**":
  - Psychology
  - Neuroscience
- So far, AI in computer science focuses more on machine's behaviors than **human-like thinking**.
  - Similar behaviors could be achieved by different mechanisms.
  - Currently, we know little about our brains.

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A HISTORY OF AI

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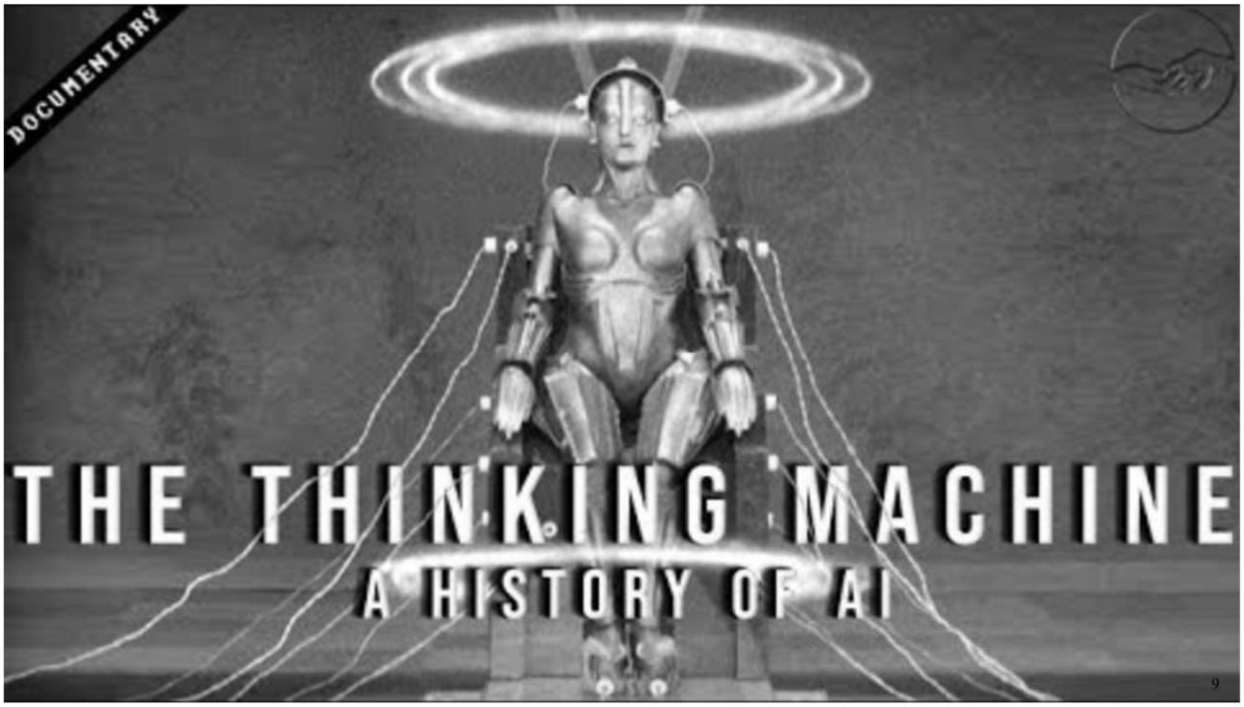

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AI Summers and Winters

- First **Summer** (1950-1966)
- First **Winter** (1967-1977)
- Second **Summer** (1978-1987)
- Second **Winter** (1988-2011)
- Third **Summer** (2012-Present)

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First AI Summer (1950-1966)

- Turing Machine (1936)
- Modern computer development during WWII
- Turing Test (1950)
- The Dartmouth workshop (1956) – a two-month workshop at Dartmouth College
  - Initiated by John McCarthy
  - Coined down the term "Artificial Intelligence"

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First AI Summer (1950-1966)

- Logic Theorist (Newell, Simon, Shaw, 1956)
  - Proves math theorems
  - Proves 38 out of 52 theorems in Principia Mathematica
- The First Computer Chess (Bernstein, 1957)
  - Can defeat an inexperienced player
- Symbolic Automatic INTegrator (Slagle, 1961)
  - Calculates (symbolic) integration
  - Solves 52 out of 54 problems in MIT's freshman final

based on **SEARCH** algorithms

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Alex Bernstein's Chess Machine

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ELIZA – The First Chatbot

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First AI Winter (1967-1977)

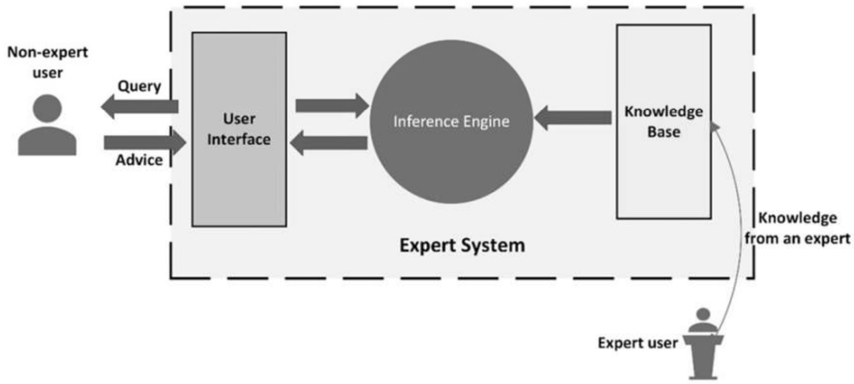
- ALPAC report (1966)
  - AI failed in machine translation between English and Russian.
- Lighthill report (1973)
  - AI research failed to address combinatorial explosion in real-world problems.
- Funding agencies (Defense Advanced Research Projects Agency, National Research Council) cutoff fundings for AI research.

https://en.wikipedia.org/wiki/AI_winter

Lessons from the First Wave of AI

- Tasks that seem easy for human may be difficult for machines.
- On the other hand, tasks that seem difficult for human might actually accomplished with simple rules.

Second AI Summer (1978-1987): Expert Systems



Represent **experts' domain knowledge** in the from of rules:

```
If [premises] then [conclusion]
```

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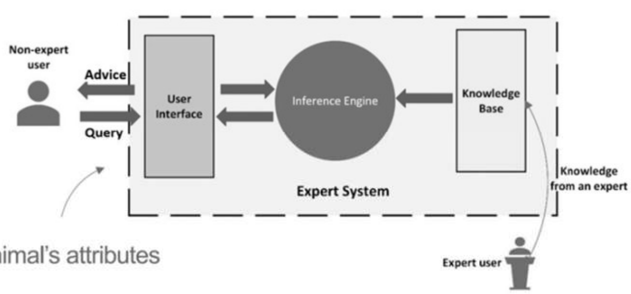
A Toy Expert System: Animal Identification



Animal's attributes

Knowledge Base

If **has_hair**, then **mammal**.
If **mammal** and **has_hooves**, then **ungulate**.
If **has_feathers**, then **bird**.
If **mammal** and **carnivore** and **has_dark_spots**, then **cheetah**.
If **mammal** and **carnivore** and **has_black_stripes**, then **tiger**.
If **bird** and **does_not_fly** and **has_long_neck**, then **ostrich**.
......

User Interface

```
File  Edit  Settings  Run  Debug  Help
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.6)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

?- go.
Does the animal have hair? yes.

Does the animal eat meat? |: no.

Does the animal have pointed teeth? |: no.

Does the animal have hooves? |: yes.

Does the animal have a long neck? |: yes.

Does the animal have long legs? |: yes.

I guess that the animal is: giraffe
true.

?- 
```

Related topics:

Logic and **Knowledge Representation**

Second AI Summer (1978-1987): Expert Systems

- DENDRAL (1968)
  - Predict molecular structure based on spectrographic data
- MYCIN (1975)
  - Diagnose blood infections
  - Better than junior doctor
- XCON (1978)
  - Select computer system components based on customer’s need
  - Save ~$25M a year
- Many companies built expert systems and software/hardware specialized for their purpose.

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Second AI Winter (1988-)

- Difficult to build and maintain expert systems for complex domains
  - Uncertainty
  - Could not learn from data
- Companies fell as they failed to meet their extravagant promises

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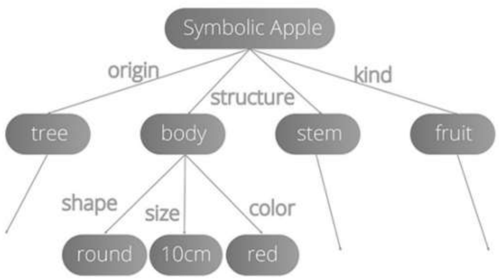
Symbolic vs. Subsymbolic AI

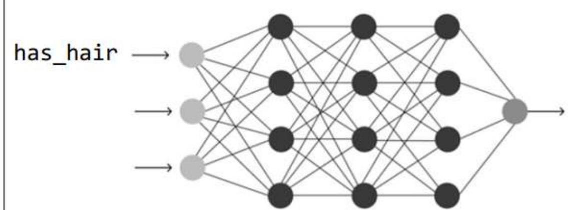
Symbolic AI

- Symbols usually directly correspond to realworld concepts or relationships.



Subsymbolic AI

- Using numerical / distributed representations for concepts and relationships.



Symbols encoded as numerical vectors, e.g., (1, 0.2, 0.1, -0.5)

Multiplying some weight matrices

Subsymbolic AI

- The “rules” (e.g., the weights in neural networks) are usually **learned from data** rather than assigned by human.
  - Less interpretable
  - But can represent more complex rules, e.g., natural languages rules
- Better in dealing with **uncertainty**
  - It hard / complex for human to write rules with uncertainty
- Learning from data → **Statistical Learning**

Statistical Learning

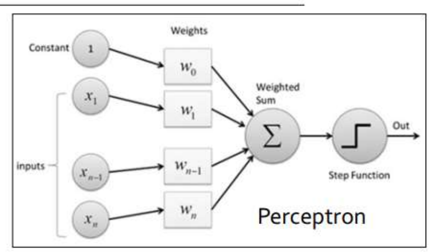
- Learning from data
- Methods
  - Linear regression (Galton, 1894)
  - Hidden Markov Model (Baum, 1960s)
  - Bayesian Network (Pearl, 1985)
  - Support Vector Machine (Cortes & Vapnik, 1995)

Modeling uncertainty

- SVM is a dominant approach in 2000-2010 for many tasks:
  - Image classification
  - Speech recognition (combined with HMM)
  - Text categorization
  - …

Artificial Neural Networks (NN)

- Perceptron (McCullough & Pitts, 1943)
  - Inspired by biological neuron
  - Can simulate logic gates
- Perceptron algorithm (Rosenblatt, 1958)
  - Learning single-layer NN



- Backpropagation (Linnainmaa, 1970; Rumelhart et al., 1986)
  - Learning multi-layer NN
- Convolutional Neural Network (Fukushima, 1980; LeCun et al., 1989)
  - Handwritten digit recognition for USPS

Third AI Summer (2012-Present): Deep Learning

- ImageNet (Fei-Fei Li et al., 2009)
  - Large-scale image dataset: 14 million images, ~20000 classes
  - Annual image recognition contest (ILSVRC) since 2010
- AlexNet (Krizhevsky, Sutskever, and Hinton, 2012)
  - Won ILSVRC by >10% margin
  - Revolutionize computer vision and spark the use of deep learning in many domains
- AlphaGo (Deepmind, 2016)
  - Won Lee Sedol (world champion of Go) 4 out of 5 games
- Transformer (Google, 2017)
  - Revolutionize natural language processing and other domains
  - Generative Pre-trained Transformer (GPT)

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Q&A

Thank you!

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